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Efficacy of individualized homeopathic medicines in intervening the progression of pre-hypertension to hypertension: A double-blind, randomized, placebo-controlled trial

Souvik Dutta[a,*], Subhasish Ganguly[a], Shyamal Kumar Mukherjee[b], Priyanka Ghosh[a], Pallavi Hazra[a], Aniket Singha Roy[a], Abdur Rahaman Shaikh[c], Sk.Swaif Ali[d], Satarupa Sadhukhan[e], Munmun Koley[f], Subhranil Saha[g]

[a] Dept. of Organon of Medicine and Homoeopathic Philosophy, D. N. De Homoeopathic Medical College and Hospital, Kolkata, West Bengal, affiliated to The West Bengal University of Health Sciences, Govt. of West Bengal, India
[b] Principal and Administrator D. N. De Homoeopathic Medical College and Hospital, Kolkata, West Bengal, affiliated to The West Bengal University of Health Sciences, Govt. of West Bengal, India
[c] Dept. of Practice of Medicine, D. N. De Homoeopathic Medical College and Hospital, Kolkata, West Bengal, affiliated to The West Bengal University of Health Sciences, Govt. of West Bengal, India
[d] Dept. of Practice of Medicine, Mahesh Bhattacharyya Homoeopathic Medical College and Hospital, Howrah, Govt. of West Bengal, affiliated to The West Bengal University of Health Sciences, Govt. of West Bengal, India
[e] Dept. of Organon of Medicine and Homoeopathic Philosophy, National Institute of Homoeopathy, Block GE, Sector III, Salt Lake, Kolkata 700106, West Bengal, India; affiliated to The West Bengal University of Health Sciences, Govt. of West Bengal, India
[f] Dept. of Organon of Medicine and Homoeopathic Philosophy, State National Homoeopathic Medical College and Hospital, Lucknow, Govt. of Uttar Pradesh, affiliated to Dr. Bhimrao Ramji Ambedkar University, Agra, Govt. of Uttar Pradesh), India
[g] Dept. of Repertory, D. N. De Homoeopathic Medical College and Hospital, Kolkata, West Bengal, affiliated to The West Bengal University of Health Sciences, Govt. of West Bengal, India

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ABSTRACT

Context: Pre-hypertension remains a significant public health challenge and appropriate intervention is required to stop its progression to hypertension and/or cardiovascular diseases.
Objective: To study the effects of individualized homeopathic medicines (IH) against placebo in intervening with the progression of pre-hypertension to hypertension.
Design: Double-blind, randomized, two parallel arms, placebo-controlled trial.
Setting: Outpatient departments of D. N. De Homoeopathic Medical College and Hospital, Kolkata, West Bengal, India.
Patients: Ninety-two patients suffering from pre-hypertension; randomized to receive either IH ($n = 46$) or identical-looking placebo ($n = 46$).
Interventions: IH or placebo in the mutual context of lifestyle modification (LSM) advices including dietary approaches to stop hypertension (DASH) and brisk exercises.
Main outcome measures: Primary – systolic and diastolic blood pressure (SBP and DBP); secondary – Measure Yourself Medical Outcome Profile version 2.0 (MYMOP-2) scores; all measured at baseline, and every month, up to 3 months.
Results: After 3 months of intervention, the number of patients having progression from pre-hypertension to hypertension between groups were similar without any significant differences in between (all $P>0.05$). Reduction in BP and MYMOP-2 scores were non-significantly higher (all $P>0.05$) in the IH group than placebo with small effect sizes. *Lycopodium clavatum*, *Thuja occidentalis* and *Natrum muriaticum* were the most frequently prescribed medicines. No harms or serious adverse events were reported from either group. Thus, there was a small, but non-significant direction of effect favoring homeopathy, which ultimately rendered the trial as inconclusive. [Trial registration: CTRI/2018/10/016,026; UTN: U1111–1221–8251]

* Corresponding author at: Dept. of Organon of Medicine and Homoeopathic Philosophy, D. N. De Homoeopathic Medical College and Hospital, 12, Gobinda Khatick Road, Tangra, Kolkata, West Bengal 700046, India.
E-mail address: drsouvikdutta@gmail.com (S. Dutta).

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Introduction

Pre-hypertension (ICD-10-CM R03.0)[1] is a sub-clinical condition, but remains a public health challenge globally. Appropriate intervention is needed to stop its progression to hypertension and other cardiovascular diseases. According to 8th Joint National Committee (JNC-8),[2] blood pressure (BP) is classified as normotensive (≤ 120/80 mm Hg), high normal blood pressure or pre-hypertensive (120−139/80−89 mm Hg) and hypertensive (≥ 140/90 mm Hg).[3] In a meta-analysis, prevalence of childhood hypertension was found to be increased with a relative increasing rate of 75% to 79% from 2000 to 2015.[4] The prevalence of pre-hypertension in Indian population was more than 45%.[5] A recent study showed that during 2011−2012, 28.2% of the US adults (≥ 20 years old) were pre-hypertensive.[6] In 2014, a study in West Bengal police personnel showed that around 42.9% police of various district of West Bengal have pre-hypertension with increased risk of cardiovascular disease conditions.[7] Another study showed that pre-hypertension was present in 41.1% of Indian urban population of Belgaum city and recognized the necessity for detecting pre-hypertension and emphasizing on dietary and lifestyle modification (LSM) to prevent further progression to hypertension and its complications.[8] Controlling pre-hypertension is basically a primary prevention managed with diet, exercise and life style modifications.[9]

Now-a-days, complementary and alternative medicine (CAM) therapies are increasingly becoming popular in various chronic diseases including hypertension.[10] Selective and integrative use of different CAM therapies has also been recommended in hypertension.[11, 12] Hypertension is a frequently encountered condition in routine homeopathy practice as well.[13] Literature search revealed 13 published trials of homeopathy in hypertension; however, none on pre-hypertension. In this trial, we hypothesized that there might be significant difference between individualized homeopathic medicines (IH) and placebo in intervening with the progression of pre-hypertension to hypertension over 3 months of intervention while both the groups received standardized recommendations for dietary and lifestyle modifications (LSM).

Materials and methods

Study design: A double-blind, randomized, placebo-controlled, two parallel arms, clinical trial was conducted in the outpatient department of D. N. De Homoeopathic Medical College and Hospital, Govt. of West Bengal, India. The study protocol was approved by the Institutional Ethical Committee (IEC) (Ref. No. DHC/Aca(PG)29/14/861/18; dated September 13, 2018) and was registered prospectively in the Clinical Trials Registry − India (CTRI/2018/10/016026; secondary identifier UTN: U1111−1221−8251). The full dissertation was submitted as the postgraduate thesis of the corresponding author to The West Bengal University of Health Sciences.

Participants: The inclusion criteria were adults aged between 18 and 65 years and of either sex, newly diagnosed subjects suffering from pre-hypertension (JNC-8 classification[2]) systolic BP (SBP) 120−139 mm of Hg and diastolic BP (DBP) 80−89 mm of Hg on at least two readings at an interval of more than one-minute,[14] new pre-hypertensive patients (those who are accidentally discovered as pre-hypertensive while complaining of some other disorders), literate, and consenting to participate. Exclusion criteria were body mass index (BMI) of 35 or higher, pregnant, puerperal and lactating women, pre-existing chronic debilitating illness, subjects on hormonal contraceptives, steroids or non-steroidal anti-inflammatory drugs (NSAIDs), any uncontrolled systemic, psychiatric diseases and life- threatening illness, self-reported immuno-compromised state, and undergoing homeopathic treatment for any chronic disease within last 6 months.

Intervention

1. Verum: In the experimental arm, indicated homeopathic medicines were administered in centesimal potencies and in individualized dosage. Each dose consisted of 6−8 globules (no. 10) of cane sugar, medicated with the indicated medicine (preserved in 90% v/v ethanol), taken orally on a clean tongue with empty stomach; dosage and repetition depending upon the individual requirement of the cases and as decided appropriate to the case or condition. Patients were advised to refrain from handling the globules or from eating, drinking, smoking or brushing teeth within 30 min of taking the globules and were asked to suck the globules rather than simply swallowing those. Dosage regime was once daily for 1−2 days, followed by known placebo once daily, up to reassessment after 1 month. Total duration of therapy was 3 months. The homeopathic medicines and sundry goods were purchased from Hahnemann Publishing Company® (HAPCO) as bulk product. Both medicines and placebo were repacked in identical glass bottles and labelled with codes (either 1 or 2, indicating either medicine or placebo), names and potencies of the medicines and was dispensed as per the coded random number chart. A single IH medicine was prescribed on each occasion, including the first consultation, considering the presenting symptom totality, clinical history, constitutional features, and repertorization by appropriate repertories using HOMPATH ZOMEO 3.0® software when required. Materia medica was consulted further for the final selection of medicine. Dose was also individualized as decided appropriate to the cases. In each individualized case, medicines, doses and repetition was decided by three homeopathic physicians. Selection was made by two homeopathic physicians, and in case of any differences in opinion, it was resolved by involvement of another homeopathic physician. Provision was kept to change the medicines or potencies and adjust the dosage in subsequent visits whenever required as per homeopathic principles. Two of the homeopathic physicians possessed master's degrees in homeopathy along with more than 20 years of teaching experience and practicing classical homeopathy. All the homeopathic physicians involved were affiliated with respective state councils.
2. Comparator: This group received identical looking placebo, indistinguishable from verum, for a period of 3 months. Each dose of placebo consisted of 4−6 globules (no.10), moistened with non-medicinal rectified spirit, and was instructed to be taken orally on a clean tongue, with empty stomach; dosage and repetition depending upon requirement of individual cases. Dosage regime was similar to verum.
3. General management: Along with medicines, all the randomized patients received standard advices on LSM including dietary counselling (DASH)[15] and brisk exercises for 30 min per day for a minimum of 5 days a week. DASH aimed at reduction in refined carbohydrates and fats (not exceeding 20 g/day), and inclusion of fiber-rich foods (whole grains, legumes, vegetables and fruits), customised according to region and culture. Compliance to the advices was assured by repeated reminders given verbally to the participants in every follow up visit by the research assistants.

Outcomes

1. Primary: Outcome assessment was done by measuring SBP and DBP using one specific Rossmax® GB series aneroid sphygmomanometer by blinded outcome assessors at baseline and every month up to 3 months. The individual was asked to sit quietly on a chair with feet on the floor for 5 min in a private, quiet setting with a comfortable room temperature. At least two measurements were made each time at an interval of more than one minute.[14]

2. Secondary: Measure Yourself Medical Outcome Profile-2 (MYMOP-2) questionnaire – an individualized, patient-rated outcome measure.[16] It is suitable for patients who present with symptoms related to their physical, emotional, or social condition. It is absolutely problem-specific, and includes daily activity and general wellbeing at the same time. It consists of 'symptom 1′ (domain 1), 'symptom 2′ (domain 2), 'activity' (domain 3), and 'wellbeing' (domain 4). It is also concerned with the 'duration of suffering', 'history of previous medication' and 'any dependence on medicines' in the initial assessment. The follow-up part is concerned with 'symptom 3′, that may have appeared in the meantime and 'any medication' for newly appeared symptom. Prior to use in this trial, the English version of the questionnaire underwent standardized forward-backward translation into Bengali and formal testing for validity and reliability, but will be reported elsewhere and beyond the scope of this paper. The Bengali version of MYMOP-2 is available from the authors on request.

Sample size: Formal effect size and subsequent sample size calculation was not possible on account of absence of any clinical trial evaluating efficacy of individualized homeopathic treatment against placebo in pre-hypertension in similar design. As both the groups were highly likely to improve from combined effects of homeopathic intervention and LSM advices, a medium effect size (w = 0.3) was assumed. In order to detect statistically significant difference between two proportions by goodness-of-fit chi-squared 2×2 contingency table, a sample size of 88 (i.e. 44 in each group) would give 80% power (minimum recommended) based on two-sided significance level of 5%. Allowing a provision for 5% drop-outs, the target sample size was further inflated to 92 (i.e. 46 in each group).

Randomization: By using StatTrek random number generator, the verum (IH) or comparator (placebo) was allocated a randomization chart. The random number chart was prepared by an independent 3rd party using simple block randomization method (9 blocks of $n = 10$; i.e., 9×10=90) and another block of size 2, thus maintaining a 1:1 ratio. Thus, an equal number of participants were randomized to the verum and to the control group.

Blinding: A double-blinding method was adopted: the patients, the treating physicians, the outcome assessors and the data entry operator, all remained blinded.

Allocation concealment: It was maintained by identically coded containers having alike amber-coloured glass vials coded as '1′ or '2′ indicating either of medicine or placebo, assigned randomly and confidentially by 3rd party. The coded randomization chart was made available to the blinded pharmacist to provide medicines accordingly from the coded vials filled with either medicated or non-medicated globules. Confidentiality was maintained strictly until the end of the trial. Randomization codes were broken at the end of the trial after the data set was frozen.

Statistical methods: The statistical analysis followed the intention-to-treat (ITT) approach; every included patient entered into the final analyses. Missing values were replaced by the last observation carried forward (LOCF) method. For comparing the distribution of socio-demographic variables between groups, unpaired t-tests were applied for continuous data (age and duration of suffering); and chi-squared (χ^2) tests for categorical data (sex, residence, risk factor, co-morbidities, marital status, education, employment status, and family income status). Differences in number of patients progressing from pre-hypertension to hypertension between groups were tested using chi-squared tests and the effect sizes were reported in terms of risk ratios (RR) and odds ratios (OR). Group differences were examined at baseline (to check comparability) and every month up to 3 months (to check efficacy) by unpaired t-tests and mixed design (split-half) two-ways repeated measures analysis of variance (ANOVA) models, and effect sizes were measured in terms of Cohen's d. Intra-group changes with time were measured using one-way repeated measures ANOVA. Two-ways analysis of covariance (ANCOVA) models were developed to detect differences in BP changes between symptomatic and asymptomatic groups, if any. P values less than 0.05 two-tailed were considered as statistically significant. SPSS®-IBM® version 20 software for Windows was used for analysis of data.

Ethical consideration: The study protocol was in accordance with the latest version of the Helsinki Declaration on Human Experimentation.[17] No invasive methodology was involved and there was no therapeutic experimentation. Prior to enrolment, each patient was provided with a patient information sheet in local vernacular Bengali detailing the objectives, methods, risks and benefits of participating, and confidentiality issues. Subsequent to that, written informed consent was obtained and then enrolled. Inter-current illness, acute exacerbation of chronic conditions, adverse or serious adverse event (s), wherever present, was recorded and treated accordingly as per homeopathic principles irrespective of assigned codes. If non-responding, then the patient was planned to be referred for conventional treatment; however, no such event occurred during the study. In this project, no new drug was experimented on, nor was any new treatment protocol adopted. Intervention was in strict adherence to classical homeopathy treatment. For ethical purpose, irrespective of codes, all the patients received standardized LSM guidelines. Approval was taken from the IEC prior to initiation and the study was performed under the constant supervision of the IEC.

Reporting of adverse events: Patients were instructed to report any harm, serious adverse event, unintended effect and undue aggravation; either over phone or directly in the outpatient department.

Trial reporting: Trial reporting complied to the Consolidated Standards for Reporting Trials (CONSORT)[18] and Reporting Data on Homeopathic Treatment (RedHot)[19] guidelines.

Results

Participant flow: Total 118 pre-hypertensive patients were screened as per the pre-specified inclusion and exclusion criteria; 26 were excluded on account of various reasons. Ninety-two patients met the eligibility criteria and were enrolled, baseline socio-demographic and outcome data were obtained, and the patients randomized subsequently into either verum or placebo. The outcome data was recorded every month up to 3 months. During the course of treatment, 16 patients dropped out (verum: 7, control: 9) and 76 completed the trial; thus, the attrition rate was 17.4% (Fig. 1).

Study duration and follow up: Total study duration was 17 months, starting from December 2018 till April 2020.

Number analyzed: All the randomized patients (n = 92) entered into the final analysis; 46 in each group.

Baseline data: Two groups were comparable as per distribution of the confounder variables – both continuous and categorical. No significant differences existed (all $P>0.05$); thus, ensuring comparability between the two groups (Table 1).

Outcomes and estimation

- ☐ Primary outcome measure – Number of patients progressing from pre-hypertension stage to hypertension were comparatively less in the IH group than placebo, but non-significant statistically at different time points (Table 2). Except DBP in the placebo group, all intra-group changes over time in the two groups were statistically significant (all $P<0.05$; one-way repeated measure ANOVA) (Table 3). Reduction in BP was higher (all $P>0.05$) in the IH group than placebo, but non-significant statistically with small effect sizes. Differences in SBP and DBP between groups were non-significant at all time points. Two-ways repeated measure ANOVA also revealed similar non-significant trends both in SBP ($F_{1,90}$=0.007, P = 0.936) and DBP ($F_{1,90}$=0.004, P = 0.835) (Table 3).

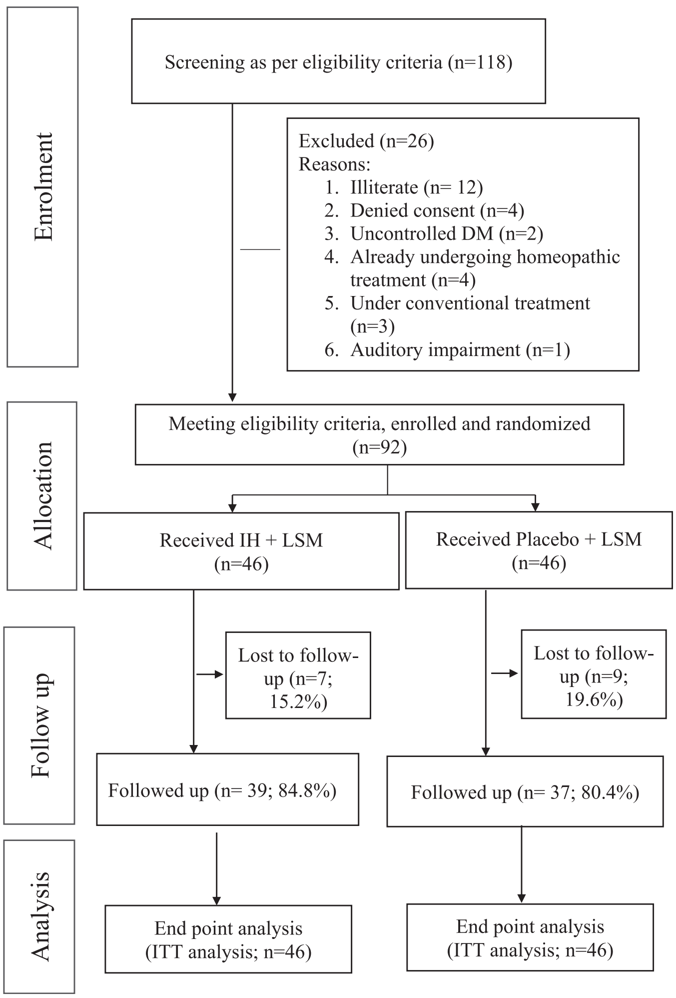

Fig. 1. CONSORT flow diagram
Abbreviations: CONSORT = Consolidated Statement for Reporting Trials; DM: Diabetes Mellitus; IH: Individualized Homeopathy; ITT: Intention to treat.

Table 1
Comparison of socio-demographic characteristics between two groups at baseline (N=92)

Features	Verum group (n=46)	Placebo group (n=46)	*P* value
Age (yrs) [a]	47.2 ± 10.1	45.3 ± 8.9	0.355
Body mass index[a]	25.0 ± 3.7	24.7 ± 3.2	0.710
Sex [b]			
Male	17 (37)	13 (28.3)	0.505
Female	29 (63)	33 (71.7)	
Residence [b]			
Rural	17 (37)	13 (28.3)	0.831
Semi-urban	2 (4.3)	3 (6.5)	
Urban	27 (58.7)	30 (65.2)	
Risk factors [b]			
Tobacco consumption	12 (26.1)	8 (17.4)	0.448
Co-morbidities [b]			
Polyarthralgia	4 (8.7)	3 (6.5)	0.936
Calcaneal spur	3 (6.5)	1 (2.2)	
Hypothyroidism	3 (6.5)	3 (6.5)	
Low backache	6 (13)	8 (17.4)	
Osteoarthritis knee	16 (34.8)	12 (26.1)	
Miscellaneous	14 (30.4)	19 (41.3)	
Marital status [b]			
Married	44 (95.7)	43 (93.5)	1.000
Single and others	2 (4.3)	3 (6.5)	
Educational status [b]			
8^{th} std. or less	40 87)	44 (95.7)	0.267
9^{th} std. or higher	6 (13)	2 (4.3)	
Employment status [b]			
Service	3 (6.5)	2 (4.3)	0.822
Business	9 (19.6)	6 (13)	
Dependent and others	34 (73.9)	38 (82.6)	
Income status [b]			
Poor	24 (52.2)	23 (50)	0.507
Middle	15 (32.6)	20 (43.5)	
Affluent	7 (15.2)	3 (6.5)	

[a] Continuous data presented as means ± standard deviations and unpaired *t*-test applied; P values less than 0.05 two-tailed were considered as statistically significant
[b] Categorical data presented as absolute values (percentages) and chi-squared test (Yates' corrected) applied; *P* values less than 0.05 two-tailed were considered as statistically significant.

Only 7 patients (verum: 2, placebo: 5) showed symptoms pertaining to HTN. No significant differences in changes in either SBP [$F_{(1, 89)}$ =0.583, *P* = 0.447, partial eta squared=0.007] or DBP [$F_{(1, 89)}$ =0.003, *P* = 0.957, partial eta squared=0.000] were noted between symptomatic and asymptomatic groups.

□ Secondary outcome measure − Intra-group changes in MYMOP-2 individual domain scores and profile scores over 1st, 2nd, and 3rd months of treatment revealed significant improvement in both the verum and the placebo groups (all $P<0.05$, one way repeated measure ANOVA), but inter-group differences in any time point could not reach statistical significance in any occasion with small effect sizes − symptom 1, symptom 2, difficulty in activity, general wellbeing and profile score (all $P>0.05$, unpaired *t*-tests). Two-ways repeated measure ANOVA also disclosed similar non-significant trends in all the MYMOP-2 subscales and profile scores (all $P>0.05$). (Table 4)

Medicines used: Total 19 different medicines were prescribed in different centesimal potencies in the two groups. *Lycopodium clavatum* (*n* = 17; 18.5%), *Thuja occidentalis* (*n* = 16; 17.4%), *Natrum muriaticum* and *Sulphur* (*n* = 12 each; 13%) were the most frequently prescribed medicines. (Table 5)

Adverse events: During the study, no events of any harm, unintended effect, homeopathic aggravation, or any serious adverse event were reported. Eighteen minor adverse events in the verum group and 20 minor adverse events in the placebo group were recorded; however, all those seemed to be independent events rather than attributable to treatment. In the verum group, few adverse events were recorded like fever, diarrhea, pain abdomen, and upper respiratory tract infections. All were treated successfully by homeopathic medicines except a case of fever where the patient took over-the-counter paracetamol 650 mg 3 tablets 6 hourly and recovered finally. In the placebo group, few adverse events like diarrhea, fever, constipation, headache, and urinary tract infection were recorded. A case of fever and another case of diarrhea took over-the-counter paracetamol 650 mg and norfloxacin 400 mg orally, completed the course of treatment and recovered. Other adverse events were successfully treated by homeopathic medications.

Discussion

In this double-blind RCT on 92 patients suffering from pre-hypertension for 3 months, a small, but non-significant direction of effect favouring homeopathy could be detected which ultimately rendered the trial as inconclusive. Number of patients progressing from pre-hypertension stage to hypertension were comparatively less in the IH group than placebo, but non-significant statistically. Mutual administration of LSM advices in both the groups and nature of the clinical condition itself might have reduced the effect size and detection of inter-group differences difficult.

The design of the trial was gold standard, that is RCT. Enrolment into the study was prospective, that is, the protocol was declared prior to enrolment of the first patient. All the collected data was

Table 2
Comparison of progression of pre-hypertension to hypertension between groups over 1, 2 and 3 months.

Months	BP	Verum (N = 46); n (%)	Placebo (N = 46); n (%)	Chi-square (Yates' corrected)	P value [(a)]	Risk ratio (95% CI)	Odds ratio (95% CI)
Month 1	Both SBP and DBP	2 (4.3)	4 (8.7)	0.178	0.673	0.5 (0.1, 2.6)	0.5 (0.1, 2.7)
	SBP only	2 (4.3)	1 (2.2)	0.000	1.000	2 (0.2, 21.3)	2 (0.2, 23.4)
	DBP only	10 (21.7)	2 (4.3)	4.696	0.030*	5 (1.2, 21.6)	6.1 (1.3, 29.7)
Month 2	Both SBP and DBP	6 (13)	6 (13)	0.096	0.757	1 (0.3, 2.9)	1 (0.3, 3.4)
	SBP only	1 (2.2)	1 (2.2)	0.511	0.475	1 (0.1, 15.5)	1 (0.1, 16.5)
	DBP only	4 (8.7)	4 (8.7)	0.137	0.711	1 (0.3, 3.8)	1 (0.2, 4.3)
Month 3	Both SBP and DBP	3 (6.5)	6 (13)	0.493	0.482	0.5 (0.1, 1.9)	0.5 (0.1, 2)
	SBP only	1 (2.2)	2 (4.3)	0.000	1.000	0.5 (0.05, 5.3)	0.5 (0.04, 5.6)
	DBP only	6 (13)	4 (8.7)	0.112	0.738	1.5 (0.4, 5)	1.6 (0.4, 6)

[(a)] Pearson's chi-square test (Yates' corrected); $^*P < 0.05$ considered as statistically significant.

converted into an analysable and reproducible master chart (soft copy, Microsoft Excel spreadsheet) where all data was extracted systematically and underwent statistical analysis subsequently. Double blinding method was adopted in the study. Allocation concealment was maintained by identically coded containers having alike vials filled with either medicated or non-medicated globules. The study was transparent in terms of declaration of protocol, and ethical conduct and reporting. Each patient was advised to follow DASH and other LSM guidelines to address ethical issues. Thus, the study conformed to every possible ethical standard. The statistical analysis has followed the ITT approach.[20, 21] Within this 17 months' time span, total 118 pre-hypertensive patients were screened as per pre-specified eligibility criteria and out of which, 92 could be enrolled, thus indicating the possibility of trials with larger sample size in the same setup in future. The inclusion/screening rate of (92/118) × 100 = 77.96% and retention rate of (76/92) × 100 = 82.60% were also promising.

During the treatment of chronic conditions, series of homeopathic medicines might be required in succession instead of single medicine to ensure a favourable response.[22] Thus the study duration might be deemed as inadequate, especially in a chronic condition like pre-hypertension. A longer follow-up might have been more appropriate; however, the ethical concerns with longer duration should be taken into consideration. Pre-hypertension is such a condition where there is scarcity of perceptible signs and symptoms, both for the patient and the physician. Studies have shown that changes in BP are related to the seasonal changes in temperature and discomfort indices.[23] BP may increase significantly during winter as compared to summer.[24] Any confounding effect of these phenomena, if present, have been balanced mutually in this trial due to its randomized, parallel arm design. Homeopathic treatment is based on signs and symptoms; so, it became difficult very often to prescribe the suitable homeopathic remedies by considering signs and symptoms of apparently healthy and mostly asymptomatic patients with pre-hypertension. The meticulous case-taking and consultation might have caused additional subjective bias, but should have been balanced by the process of randomization. On account of absence of any published trial of similar design, our sample size calculation was based on assumption

Table 3
Comparison of the primary outcome measure systolic and diastolic blood pressure at baseline and after 1, 2 and 3 months (N=92; Verum group: 46, Placebo group: 46)

Blood pressure	Baseline :Mean ± SD	After 1 month: Mean ± SD	After 2 months: Mean ± SD	After 3 months: Mean ± SD	Wilks' lambda	Partial eta squared	$F_{3, 43}$	P value [(c)]
SBP								
Verum group	135.2 ± 3.4	129.1 ± 6.9	129.6 ± 9.8	127.7 ± 8.9	0.524	0.476	13.025	<0.001***
Placebo group	133.8 ± 4.0	129.5 ± 8.2	128.9 ± 9.7	129.1 ± 8.9	0.631	0.369	8.382	<0.001***
Mean group difference ± SE	1.4 ± 0.8	-0.4 ± 1.6	0.7 ± 2.0	-1.4 ± 1.8				
95% CI	-0.1, 3.0	-3.5, 2.7	-3.3, 4.7	-5.0, 2.3				
t_{90}	1.841	-0.248	0.353	-0.725				
P value [(a)]	0.069	0.805	0.725	0.470				
Effect size (Cohen's d)	–	0.053	0.072	0.157				
Two-ways repeated measure ANOVA								
-$F_{1, 90}$				0.007				
-Partial eta squared				0.000				
-P value [(b)]				0.936				
DBP								
Verum group	85.9 ± 2.0	84.4 ± 4.6	83.9 ± 6.3	82.0 ± 6.2	0.700	0.300	6.133	0.001**
Placebo group	85.7 ± 2.1	84.1 ± 5.5	83.7 ± 6.4	83.4 ± 6.3	0.867	0.133	2.200	0.102
Mean group difference ± SE	0.2 ± 0.4	0.3 ± 1.1	0.2 ± 1.3	-1.4 ± 1.3				
95% CI	-0.7, 1.0	-1.8, 2.4	-2.4, 2.9	-4.0, 1.1				
t_{90}	0.411	0.327	0.180	-1.102				
P value [(a)]	0.682	0.744	0.857	0.273				
Effect size (Cohen's d)	–	0.059	0.031	0.224				
Two-ways repeated measure ANOVA								
-$F_{1, 90}$				0.004				
-Partial eta squared				0.000				
-P value [(b)]				0.835				

SBP: systolic blood pressure; DBP: diastolic blood pressure; SD: standard deviation; CI: confidence interval; SE: standard error.
[(a)] Unpaired t tests; t_{90}: t score at 90 degrees of freedom.
[(b)] Two-ways repeated measure ANOVA; P value [(a)]: inter-group differences detected by unpaired t-tests; P value [(b)]: inter-group differences detected by two-ways repeated measure ANOVA models; P value [(c)]: intra-group changes detected by one way repeated measure ANOVA; Lambda is the measure of the percentage variance in dependent variables not explained by differences in levels of the independent variable. The null hypothesis is rejected when Wilks' lambda is close to zero; Eta is the measure of effect size for use in ANOVA and analogous to R^2 from multiple linear regressions; $^{**}P < 0.01$, $^{***}P < 0.001$.

Table 4
Comparison of the secondary outcome MYMOP-2 scores at baseline and over 1, 2 and 3 months (N=92; Verum group: 46, Placebo group: 46)

MYMOP-2 scores	Baseline: Mean ± SD	After 1 month: Mean ± SD	After 2 months: Mean ± SD	After 3 months: Mean ± SD	Wilks' lambda	Partial eta squared	$F_{3, 43}$	P value [c]
MYMOP-2 Symptom 1								
Verum group	3.0 ± 1.3	2.4 ± 1.4	2.2 ± 1.5	1.8 ± 1.4	0.465	0.535	16.508	<0.001***
Placebo group	3.3 ± 1.4	2.6 ± 1.5	2.4 ± 1.6	1.9 ± 1.6	0.435	0.565	18.619	<0.001***
Mean group difference ± SE	-0.3 ± 0.3	-0.2 ± 0.3	-0.2 ± 0.3	-0.1 ± 0.3				
95% CI	-0.9, 0.2	-0.8, 0.4	-0.8, 0.4	-0.7, 0.5				
t_{90}	-1.239	-0.722	-0.547	-0.341				
P value [a]	0.218	0.472	0.586	0.734				
Effect size (Cohen's d)	–	0.138	0.129	0.067				
Two-ways repeated measure ANOVA								
-$F_{1, 90}$				0.567				
-Partial eta squared				0.006				
-P value [b]				0.453				
MYMOP-2 Symptom 2								
Verum group	1.3 ± 1.6	0.9 ± 1.5	0.9 ± 1.3	0.5 ± 0.9	0.658	0.342	7.446	<0.001***
Placebo group	1.4 ± 1.7	1.3 ± 1.6	1.1 ± 1.5	1.0 ± 1.3	0.764	0.236	4.428	0.008**
Mean group difference ± SE	-0.1 ± 0.3	-0.4 ± 0.3	-0.2 ± 0.3	-0.5 ± 0.2				
95% CI	-0.8, 0.6	-1.0, 0.3	-0.8, 0.3	-0.9, 0.02				
t_{90}	-0.247	-1.151	0.219	-1.908				
P value [a]	0.805	0.253	0.416	0.060				
Effect size (Cohen's d)	–	0.258	0.142	0.447				
Two-ways repeated measure ANOVA								
-$F_{1, 90}$				1.067				
-Partial eta squared				0.012				
-P value [b]				0.304				
MYMOP-2 Difficulty in activity								
Verum group	3.0 ± 1.2	2.4 ± 1.2	2.1 ± 1.2	1.8 ± 1.3	0.476	0.524	15.783	<0.001***
Placebo group	3.2 ± 1.4	2.6 ± 1.4	2.3 ± 1.3	2.1 ± 1.4	0.456	0.544	17.067	<0.001***
Mean group difference ± SE	-0.2 ± 0.3	-0.2 ± 0.3	-0.2 ± 0.3	-0.3 ± 0.3				
95% CI	-0.8, 0.3	-0.7, 0.4	-0.7, 0.3	-0.9, 0.3				
t_{90}	-0.874	-0.691	-0.814	-1.070				
P value [a]	0.384	0.491	0.418	0.287				
Effect size (Cohen's d)	–	0.153	0.160	0.222				
Two-ways repeated measure ANOVA								
-$F_{1, 90}$				0.880				
-Partial eta squared				0.010				
-P value [b]				0.351				
MYMOP-2 General wellbeing								
Verum group	2.9 ± 1.1	2.4 ± 1.2	2.1 ± 1.2	1.8 ± 1.2	0.470	0.530	16.168	<0.001***
Placebo group	3.3 ± 1.4	2.6 ± 1.4	2.4 ± 1.3	2.1 ± 1.3	0.447	0.553	17.746	<0.001***
Mean group difference ± SE	0.4 ± 0.3	-0.2 ± 0.3	-0.3 ± 0.3	-0.3 ± 0.3				
95% CI	-0.9, 0.2	-0.7, 0.3	-0.9, 0.1	-0.8, 0.2				
t_{90}	-1.138	-0.727	-1.401	-1.251				
P value [a]	0.191	0.469	0.165	0.214				
Effect size (Cohen's d)	–	0.153	0.240	0.240				
Two-ways repeated measure ANOVA								
-$F_{1, 90}$				1.645				
-Partial eta squared				0.018				
-P value [b]				0.203				
MYMOP-2 Profile score								
Verum group	2.5 ± 1.0	2.0 ± 1.0	1.8 ± 1.0	1.5 ± 1.0	0.447	0.553	17.720	<0.001***
Placebo group	2.8 ± 1.2	2.2 ± 1.2	2.1 ± 1.2	1.8 ± 1.2	0.392	0.608	22.248	<0.001***
Mean group difference ± SE	-0.3 ± 0.2	-0.2 ± 0.2	-0.3 ± 0.2	-0.3 ± 0.2				
95% CI	-0.7, 0.2	-0.7, 0.2	-0.7, 0.2	-0.7, 0.2				
t_{90}	-1.107	-1.204	-1.059	-1.291				
P value [a]	0.271	0.309	0.293	0.200				
Effect size (Cohen's d)	–	0.181	0.272	0.272				
Two-ways repeated measure ANOVA								
-$F_{1, 90}$				1.450				
-Partial eta squared				0.016				
-P value [b]				0.232				

MYMOP-2: Measure yourself medical outcome profile version 2.0; SD: standard deviation; CI: confidence interval; SE: standard error.

[a] Unpaired t tests; t_{90}: t score at 90 degrees of freedom.

[b] One way repeated measure analysis of variance; $P < 0.05$ considered as statistically significant; P value [a]: inter-group differences detected by unpaired t-tests; P value [b]: inter-group differences detected by two-ways repeated measure ANOVA models.

[C] Intra-group changes detected by one way repeated measure ANOVA; Lambda is the measure of the percentage variance in dependent variables not explained by differences in levels of the independent variable. The null hypothesis is rejected when Wilks' lambda is close to zero; Eta is the measure of effect size for use in ANOVA and analogous to R^2 from multiple linear regressions; $^{**}P < 0.01$, $^{***}P < 0.001$.

Table 5
Alphabetical list of medicines prescribed in the two groups at baseline.

Name of the medicines	Total (N = 92); n (%)	Verum group (n = 46); n (%)	Placebo group (n = 46); n (%)	P value [(a)]
1. *Aconitum napellus*	1 (1.1)	1 (2.2)	–	–
2. *Argentum nitricum*	2 (2.2)	–	2 (4.3)	–
3. *Arsenicum album*	3 (3.3)	–	3 (6.5)	–
4. *Bacillinum*	3 (3.3)	1 (2.2)	2 (4.3)	1.000
5. *Borax venenata*	2 (2.2)	1 (2.2)	1 (2.2)	0.475
6. *Calcarea carbonica*	2 (2.2)	–	2 (4.3)	–
7. *Causticum*	3 (3.3)	–	3 (6.5)	–
8. *Gelsemium sempervirens*	1 (1.1)	–	1 (2.2)	–
9. *Ignatia amara*	3 (3.3)	–	3 (6.5)	–
10. *Lycopodium clavatum*	17 (18.5)	12 (26.1)	5 (10.9)	0.107
11. *Medorrhinum*	2 (2.2)	1 (2.2)	1 (2.2)	0.475
12. *Natrum muriaticum*	12 (13)	6 (13)	6 (13)	0.757
13. *Natrum sulphuricum*	5 (5.4)	5 (10.9)	–	–
14. *Nitricum acidum*	1 (1.1)	1 (2.2)	–	–
15. *Pulsatilla nigricans*	5 (5.4)	3 (6.5)	2 (4.3)	1.000
16. *Sepia succus*	1 (1.1)	1 (2.2)	–	–
17. *Staphysagria*	1 (1.1)	–	1 (2.2)	–
18. *Sulfur*	12 (13)	5 (10.9)	7 (15.2)	0.757
19. *Thuja occidentalis*	16 (17.4)	9 (19.6)	7 (15.2)	0.783

[(a)] Pearson's chi-square test (Yates' corrected) or Fisher's exact test; $P < 0.05$ considered as statistically significant.

of effect sizes, especially in the context of confounders like LSM advices. Given the estimated effect size of 0.157 as detected on SBP after 3 months of intervention, in order to detect statistically significant differences between two independent means of SBP using the unpaired t-test, any further replication study would require 2×638 patients to provide with 80% power based on a two-sided significance level of 5% by maintaining 1:1 allocation. Keeping DBP as the primary outcome, desired sample size would be 2×314. Provision of attrition rate of 17.4% as reported in this trial will further inflate the target sample size up to 1498 and 738 keeping SBP and DBP as the primary outcomes respectively. Another important limitation was that only verbal compliance to the advised interventions (medicines and LSM) could be assured in our study; still, compliance bias, if any, randomization would have mitigated the effects mutually in both arms. Future trials should keep provisions for subject registries and medication adherence technologies to account for the increased variance and reduced magnitude of treatment effects owing to non-adherence.[25] Absence of any pre-hypertension specific quality of life measure compelled to adopt a patient-reported generic questionnaire like MYMOP-2. Use of particular adverse event reporting criteria or registries would have improved the reporting further.

A literature search revealed two cross-over studies of homeopathic medicines in hypertension.[26, 27] Several studies were conducted where the non-individualized prescriptions were made.[28-32] Both individualized and fixed medicines were used in one study.[33] One observational study was also identified.[34] There were three double-blind, placebo-controlled RCTs.[35-38] In one double-blind RCT on 68 patients suffering from mild to moderate hypertension, homeopathic treatment was found to be successful in 82.2% of patients as compared to 56.7% in placebo.[35] Significant reduction of BP was observed in another RCT on 129 older adults suffering from hypertension where IH was used as an adjuvant to standard therapy.[35] In another double-blind RCT on 150 patients, IH produced a significantly different hypotensive effect than placebo after 3 and 6 months of intervention.[37] In a recently published pilot trial on 68 patients suffering from stage I hypertension, group differences were seemingly higher in the IH group than in the placebo group; however the differences were non-significant statistically.[38] And until recently, a single-blind RCT on stage I HTN has also been published.[39] However, none of the published studies were on pre-hypertension and ours was the first ever conducted exploratory, double-blind RCT of IH in pre-hypertension.

Future researches may be aimed at exploring the effectiveness of IH, LSM and integrated IH plus LSM in pre-hypertension using pragmatic study designs.

Conclusion

In this double-blind RCT, there was a small, but non-significant direction of effect favouring homeopathy, which ultimately rendered the trial as inconclusive. Any effect of IH differentiable from placebo in preventing progression from pre-hypertension to hypertension could not be detected clearly; hence null hypothesis of no difference could not be rejected.

Authors' contribution

S.D., S.G., S.K.M., P.G., P.H., A.S.R. and A.R.S. contributed to the literature search, study concept, conducting the trial, data collection, data evaluation and drafting the manuscript. SSA, S. Sadhukhan, M.K and S. Saha contributed to study design, data interpretation, statistical analysis and drafting of the manuscript. All the authors reviewed and approved the final manuscript.

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Declaration of Competing Interest

None declared.

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